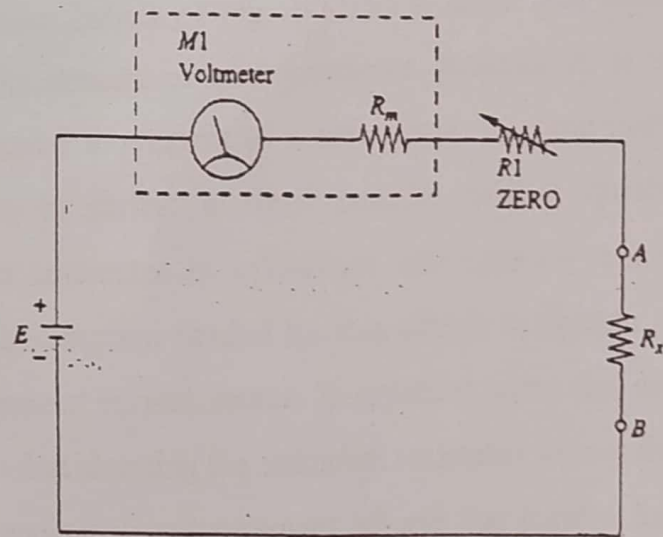


electronic voltmeter suffers from input biasing due to electromagnetic interference. People who work on radio transmitters find this problem quite often. The passive meter circuitry of the volt-ohm-millimeter is far more immune to EMI than even most of the best active meters. Unfortunately, the kind of filtering methods that are used to eliminate EMI also sometimes distort waveforms being measured.

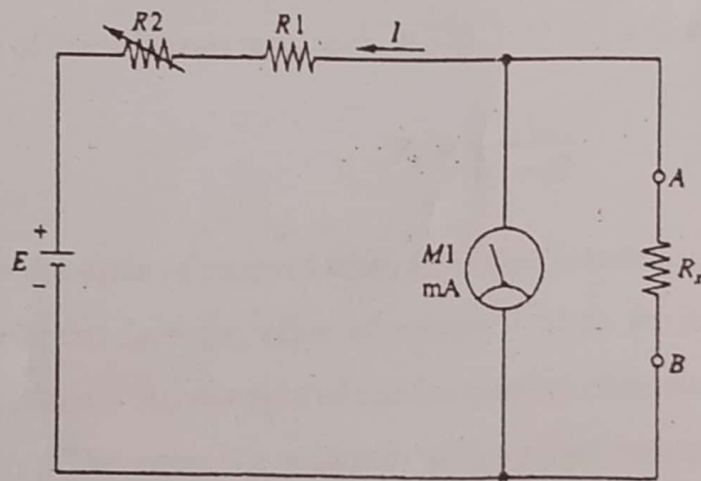
DC ANALOG OHMMETERS

Electrical resistance is measured using an instrument called an **ohmmeter**. According to Ohm's law, the current in a circuit is directly proportional to the applied voltage (E) and inversely proportional to the resistance (R) of the circuit; that is, $I = E/R$. That relationship can be used to find an unknown resistance (R_x), if a stable voltage is available. The circuit for a simple series **ohmmeter** is shown in Fig. 1-11(a). There are three resistances in this circuit: R_m is the internal resistance of the voltmeter (M), R_1 is the resistance of a zero calibration control potentiometer, and R_x is the unknown resistance.

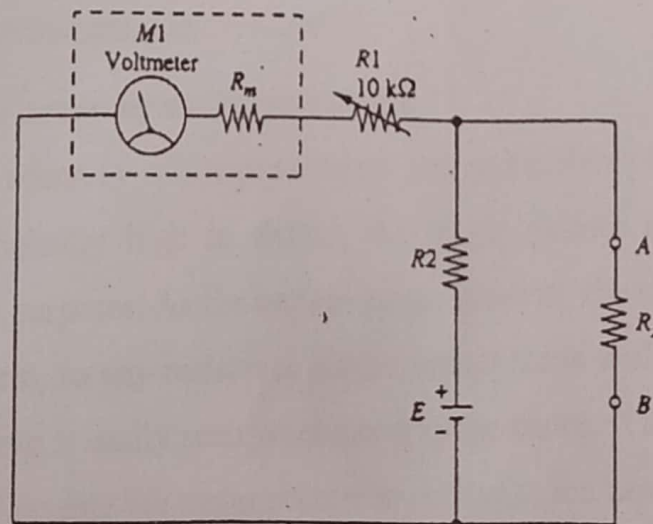
The scale for ohmmeters is reversed in direction from the voltage and current scales. That is, the highest values are on the left side of the scale, while the lowest values are on the right. In the ohmmeter of Fig. 1-11(a), the zero point occurs when maximum current flows (which puts the meter on the right side), that is, when terminals A and B are shorted together. Typical operation of the analog ohmmeter requires that the leads be shorted together with the pointers zeroed with the calibration control. On modern digital voltmeters, by the way, the zeroing is automatic. A limitation of the series ohmmeter circuit is that it does not work well at low resistances ($R_x < 100\Omega$). For lower resistance ranges the parallel ohmmeter, also called the



(a)



(b)



(c)

Fig. (1-11) Circuits for (a) series ohmmeter, (b) parallel (short) ohmmeter, (c) series-parallel ohmmeter.

shunt ohmmeter, circuit of Fig. 1-11(b) is used. The meter movement in this case is a milliammeter or microammeter. Resistor R_I is used to limit current flow in the meter movement to a safe value. In some instances, R_I consists of two resistors in series: a fixed resistor and a variable resistor (e.g., a potentiometer connected as a rheostat). The variable resistor acts as the ZERO control. An assumption needed for this circuit to operate successfully is that it be a near-constant current source. In practical terms this assumption means that $RI \gg R_m$ so that shunting the unknown resistance across the meter resistance during the resistance measurement affects the current only a small amount. This requirement usually means that $R_I = 100 \times R_m$ or higher. The value of the unknown resistance, in Fig. 1-11 (b) is found from

$$R_x = \frac{I_2 R_m}{I_1 - I_2} \quad (1-11)$$

where I_1 = the value of current I when R_x is disconnected (i.e., when terminals A and B are open) I_2 = the value of current I when R_x is connected between terminals A and B . An example of a series-parallel ohmmeter circuit is shown in Fig. 1-11(c). The meter is a voltmeter with a series resistor (R_1) set to the full-scale voltage. The Voltmeter measures the voltage drop across points A - B , with R_2 being permanently in the Circuit.

Ohmmeter accuracy vs. Battery aging

The accuracy of the ohmmeters presented above depends on the voltage being sufficiently high to deflect the meter pointer to full scale for zero-calibration purposes. As the battery ages, however, the meter will not be able to come to zero, so any resistance measurement made will be in substantial error. This problem is easily seen when zeroing the meter: if the ZERO control is not capable of causing the meter pointer to deflect to the zero marker, then there is a problem, and the battery must be changed.

Somewhat more subtle, however, is the stage immediately prior to the point where the battery weakness is obvious on the zero scale. As the battery deteriorates, its internal resistance increases, and this resistance can cause an error in the reading produced. It has the effect, for example, of increasing the apparent internal resistance of the meter movement by an amount that was not taken into account when the dial scale was calibrated.

For either problem the solution is to keep fresh batteries in the instrument at all times. Even if the meter is not used, the batteries should be refreshed at least every six months or so, or more often if the manufacturer so recommends.

DC and AC Deflection Meter Movements

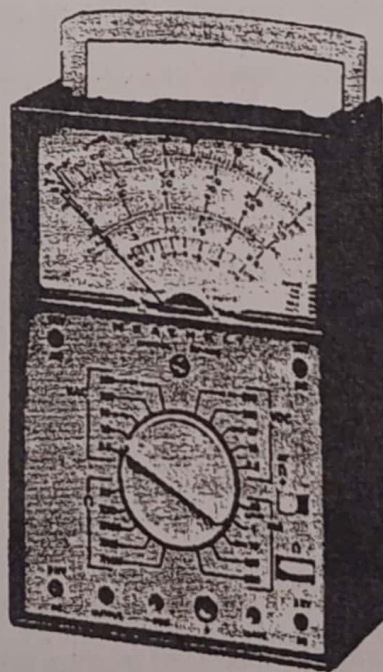


Fig. (1-12) Analog multimeter.

ANALOG MULTIMETERS

A **multimeter** (Fig. 1-12) is an instrument that has several switch-selectable voltage, current, and resistance scales. The main selector switch connects the shunt, multiplier, or range resistors, as required. Most instruments also include a rectifier to allow reading ac values (see Section 1-15).

When using a multimeter, you must follow the same rules as in individual current meters, voltmeters, and ohmmeters, so be very careful. It is all too easy to change the selector switch to a different function without changing the way the meter is connected into the circuit—for example, going from a voltage function to a current function without changing the connection from parallel to series. A typical multimeter might have dc voltage ranges from 0.5 V dc full scale to 1500 V dc full scale, so there is always a possibility of destruction of the instrument if the connection rules are not followed.

It is good laboratory practice to always store a multimeter with the switch in the OFF position; if the meter is a passive volt-ohm-milliammeter (YOM) with no OFF position, then store it with the selector set to the highest DC VOLTS range position. When using a YOM, begin with the highest range current or voltage (as needed), and work down to a range that gives a readable deflection.

ANALOG ALTERNATING CURRENT (AC)

Deflection-Type Meters

Neither type of basic dc PMMC meter movement will correctly indicate the value of an alternating current (ac), and indeed either might be destroyed by applying ac. There are, however, four basic types of meter movements that will indicate ac values: thermocouple, hot-wire, electrodynamometer, and iron-vane. We can also make a dc meter movement (d'Arsonval or taut-band) read ac values if a rectifier or integrated circuit rms-to-dc converter is used between the ac input and the dc meter movement.

WHICH AC VALUE IS BEING MEASURED?

The measurement of voltage and current values in dc circuits is straightforward because these values do not vary. In ac circuits, however, the situation is confusing because the ac varies cyclically, there are different values recognized, and all of the values can legitimately be measured. Adding to the confusion is the fact that different types of ac meters inherently display one or the other of the different ac values.

Consider the sine wave in Fig. 1-13. Assuming that there is neither dc offset nor harmonic content (i.e., a "pure" sine wave that is symmetrical on both half-cycles), several different values can be measured (let's consider only the voltage values, for the current values are of the same sort): peak voltage (E_p), peak-to-peak voltage (E_{p-p}), root mean square voltage (E_{rms}), and average voltage (E_{av}).

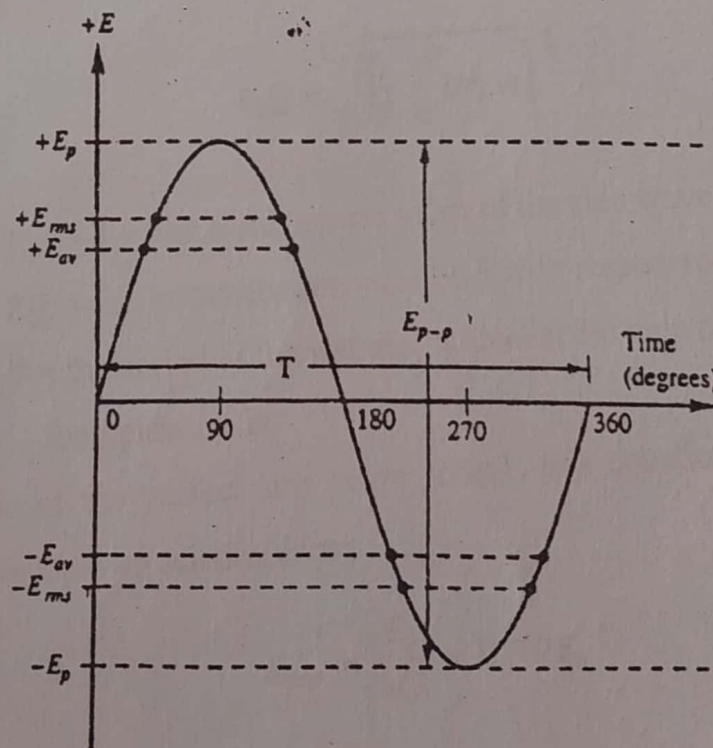


Fig. (1-13) Sine wave relationships

The peak voltage (E_p) is the voltage from the zero baseline to the peak of either half of the sine wave; the positive peak is $+E_p$, while the negative peak is $-E_p$. The peak-to-peak voltage is the magnitude of the voltage between the peak voltages:

$$E_{p-p} = |+E_p| + |-E_p| \quad (1-13)$$

In the case of a perfect sine wave, with neither a dc offset nor any significant harmonic distortion, the peak-to-peak voltage is equal to twice either the positive or the negative peak voltage.

The root mean square voltage (E_{rms}) is also called the effective voltage because it relates to the amount of power that can be delivered to a load. The rms value is equivalent to the dc voltage that will produce the same amount of heating in a resistive load. For any ac waveform, the rms value can be computed from

$$E_{rms} = \sqrt{\left[\frac{1}{T} \int_0^T E(t) dt \right]^2} \quad (1-14)$$

where E_{rms} = the root mean square value of the sine wave voltage
 $E(t)$ = the instantaneous value of E with respect to time (1-14)
 T = the period of the waveform, that is, the time required for one complete cycle

For the case of the perfect sine wave (*only*); this equation evaluates to the following well known relationships:

$$E_{rms} = \frac{E_p}{\sqrt{2}} = 0.707E_p \quad (1-15)$$

or

$$E_{rms} = \frac{\sqrt{2}E_p}{2} \quad (1-16)$$

From these relationships we can also state that the peak voltage E_p is E_{rms} and that the peak-to-peak voltage E_{p-p} is $2.828E_{rms}$.

The average value of the sine wave over the entire cycle is zero ($t_1 + t_2$), but over the half-cycle it is

$$E_{av} = \frac{2E_p}{\pi} \quad (1-17)$$

or

$$E_{av} \approx 0.637E_p \quad (1-18)$$

The half-wave average value is not much use in most cases, but in the case of some ac voltmeters it can be of great importance in interpreting readings.

For waveforms other than the sine wave, the equation will evaluate differently than the sine wave case. This fact becomes important when dealing with instruments that either are calibrated for the sine wave (the common case) or are inherently read as some other value than what is important for the waveform at hand.

For example, consider the perfectly symmetrical square wave of Fig. 1-14. This waveform possesses both baseline symmetry (i.e., the positive and negative excursions are of the same magnitude and are mirror images of each other) and left-right symmetry (i.e., the left and right halves of both the positive and the negative excursions are mirror images of each other). The half-wave average is $E_p/2$

All continuous waveforms can be broken into a mathematical expression of a fundamental frequency F equal to $1/(t_1 + t_2)$ plus sine and cosine harmonics of F . This is called the Fourier series of the signal. The Fourier series is not just a mathematical description of the signal. If you employ a device called a spectrum analyzer, then you will see an amplitude-vs.-frequency component plot of the Fourier series of the applied waveform.

The main difference between signals of differing wave shapes is the particular set of harmonics present, their relative amplitudes, and their relative phase relationships. The undistorted sine wave has only the fundamental component, and no harmonics, while a square wave has a (theoretically) infinite number of even harmonics. The symmetries discussed above tell us something about the way the meter performs when measuring those signals. For the case of the square wave, the positive and negative peaks are equal, as are the durations of those peaks, so over one complete cycle ($t_1 + t_2$) the voltage averages out to zero.

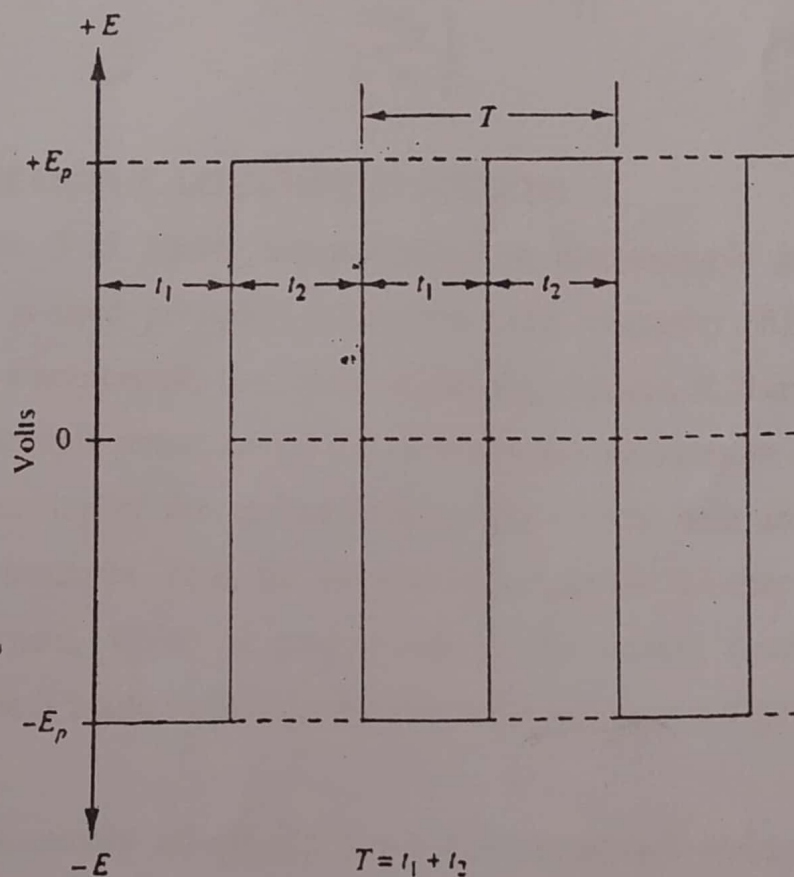


Fig. (1-14) Symmetrical square wave with baseline symmetry and left-right symmetry.

Form Factor (A)

The form factor (A) of a periodic waveform is the ratio of its rms value to its half-wave average value:

$$\Lambda = \frac{V_{rms}}{V_{av}} \quad (1-19)$$

For the case of the sine wave, the value of the form factor is

$$\Lambda = \frac{\left[\frac{\sqrt{2}E_p}{2} \right]}{\left[\frac{2E_p}{\pi} \right]} = 1.11 \quad (1-20)$$

THERMOCOUPLE AC CURRENT METERS

Figure 3-15 shows schematically the thermocouple ammeter. This instrument is used primarily to measure radio frequency (RF) currents. It consists of a nonreactive (i.e., resistive) heating element, R , thermally coupled to a thermocouple sensor device (i.e., a sensor that produces an output voltage that is a function of the applied temperature), so that both are in the same thermal environment. Thus, the thermocouple measures the temperature of the heating element, which is proportional to the current flowing in it. A thermocouple is made by joining two dissimilar metals in a V junction (Fig. 1-15).

This assembly effectively forms a temperature-to-voltage transducer; when the thermocouple junction is heated, a voltage appears across terminals A and B . This phenomenon, called the Seebeck effect, occurs because of the different *work functions* of the two metals (which is why they must be *dissimilar*, i.e., have different work functions) forming the thermocouple. Although the voltage is quite nonlinear over its entire range, it is reasonably linear over the portion of its entire range that is used by the meter. Note,

however, that zero-referenced thermocouple meters are not linear in the low end of their scale. It is necessary to select an instrument that will deflect to the upper half of its full-scale range for the current expected in the circuit.

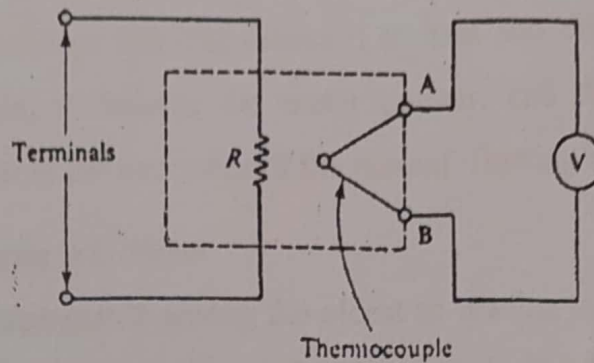


Fig. (1-15) Thermocouple meter circuit.

The alternating current being measured flows through the heater resistance R and causes it to rise in temperature, thereby causing the thermocouple output voltage to rise.

A voltmeter connected across terminals A and B measures this millivolt-level potential, but its scale is calibrated in amperes or milliamperes. This operation tells us something of the nature of the thermocouple ammeter measurements: they inherently represent the *rms value of the alternating current being measured*.

Almost any ac meter can be used to measure the rms value of $\sim$ sine wave because of the well-known relationships given in Section 1.16. But on nonsinusoidal waveforms these relationships are sometimes difficult to predict. Only an inherently rms-reading instrument such as the thermocouple ammeter will read the rms value of such a wave-form.

HOT-WIRE AMMETERS

Figure 1-16 shows the hot-wire ac meter movement. A resistance wire is stretched between two terminals and is pulled taut by a tension spring. The pointer is mounted to a pivot and is tied to the string and the spring. A current passing through the resistance element causes it to heat and thereby expand. When the wire expands, it deflects the meter pointer, and the amount of deflection is proportional to the rms value of the current flowing in the wire.

The Electrodynamometer AC Meter

The electrodynamicometer is among the oldest ac-reading instruments still in common use. It was invented by the Siemens brothers in Germany following the development of a similar device by Weber in 1843. By 1910 the configuration of the present instruments had been determined, and these designs are still widely used today (especially in the electrical power industry).

The basic electrodynamicometer circuit is shown in Fig. 1-17(a). This type of meter differs from the PMMC types in that it creates its own magnetic field from the current flowing in the windings of coils $L1$ through $L3$. Coils $L1$ and $L2$ are stationary, while coil

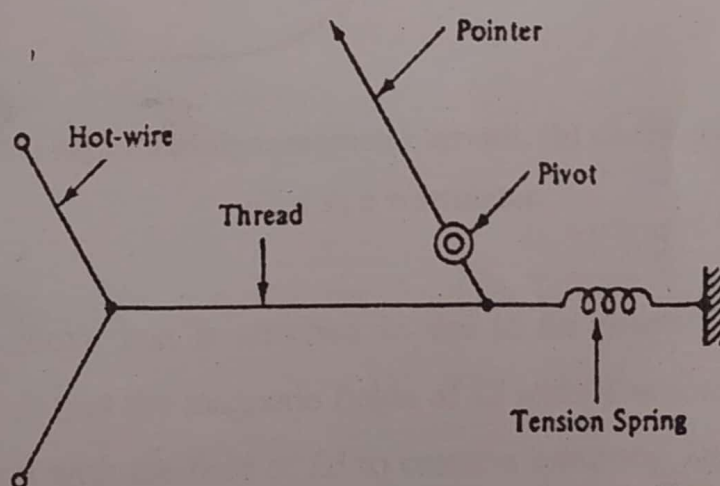


Fig. (1-16) Hot-wire ac meter movement

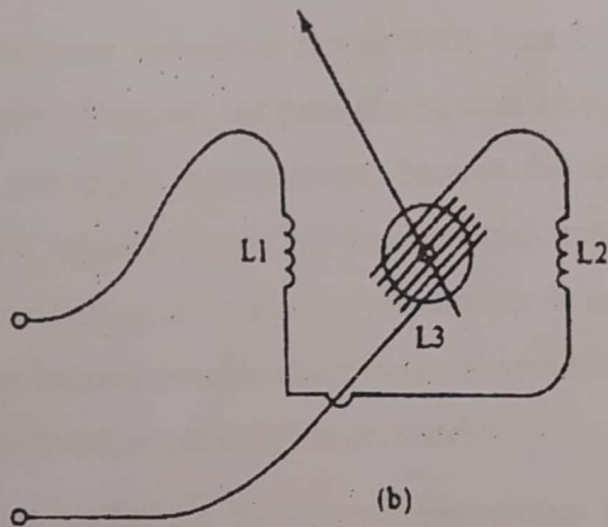
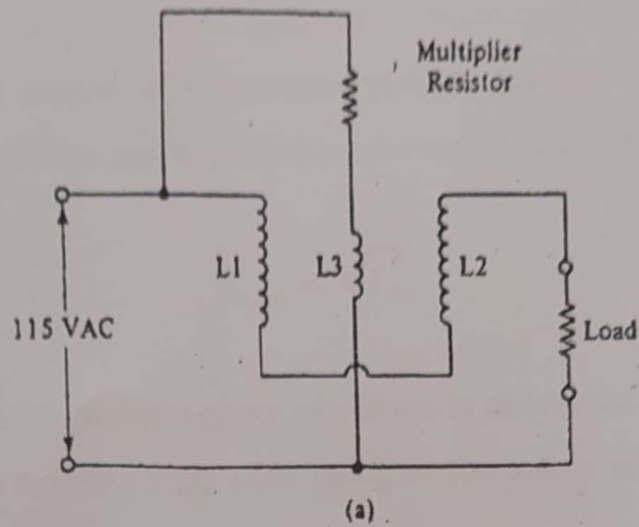


Fig. (1-17) (a) Electrodynamic circuit, (b) electrodynamicometer used as a wattmeter.

$L3$ is free to move and is attached to the meter pointer. The *sense* of the windings is such that the magnetic fields of $L1$ and $L2$ will reinforce each other and will interact with the field of $L3$ to create a rotational force on coil $L3$. The movable coil, $L3$, rotates an amount that is proportional to the strength of the current flowing in the three series-connected coils.

The electrodynamicometer is a square-law device. That is, the pointer deflects an amount that is proportional to the *square* of the *average* current flowing in its coils [i.e., $(I_{av})^2$]. In most electrodynamicometers used to measure current (some are connected to measure power), the dial scale is actually calibrated in terms of rms current, which in sine waves is defined as

$$E_{rms} = \sqrt{i_{av}^2} \quad (1-21)$$

It takes a considerable amount of current to deflect the pointer, so most electrodynamicometers have a full-scale rating of 100 mA or more, and there are many in the 0-to- 10-A range.

Electrodynamicometer voltmeters can be built from a multiplier resistor and a 100-mA meter movement, but these can be used only in low-impedance applications (e.g., power line measurements) because the sensitivity is on the order of only 10 Ω/V , which is far too low for electronic circuit applications. Extending the current range of an electrodynamicometer is usually done with a current transformer because suitable shunt resistors would be very low valued, and that renders them difficult to calibrate accurately.

One principal application for electrodynamicometers is in the wattmeter used in 60-Hz ac power line systems. An example is shown in Fig. 1-17 (b). The stationary coils are designed to carry a heavy current and are connected in series with the load. The movable coil; on the other hand, is wound of a much finer gage of wire, so it will carry a much smaller current (100 mA or less). The small coil ($L3$) is connected across the load and serves as the voltage-measuring element. Coil $L3$ may or may not have a multiplier resistor connected, depending on the design and intended application. When there is no multiplier resistor, the resistance of $L3$ must be sufficient to limit the coil current to a safe value.

The deflection of the pointer connected to $L3$ is proportional to the product of the volts times amperes ($V \times A$), which is given in watts (units of power in nonreactive circuits).

IRON-VANE METER MOVEMENTS

The iron-vane meter movement (Fig. ; -18) consists of a coil of wire to carry the current being measured and two soft iron vanes: one vane is movable, the other stationary. A pointer is attached to the movable vane. When current flows through the coil, it magnetizes the two vanes to the same magnetic polarity, and since like poles repel each other, the movable vane tends to deflect an amount that is proportional to the rms value of the current.

At ac power main frequencies (50 Hz in Europe, 60 Hz in the United States), the iron-vane meter produces accurate readings, but as the frequency of the alternating current increases, hysteresis and eddy current losses tend to produce nonlinearity problems.

At de there is another type of error: residual magnetism of the vanes by the direct current flowing in the coil. Direct current flows in only one direction, so the vanes will polarize in that direction. Alternating current, on the other hand, reverses direction every half-cycle, so the magnetic field is continually changing, thereby eliminating the buildup of a residual magnetic field in the vanes.

RECTIFIER-BASED ac METERS

Most of the ac-reading meters that you will encounter will be of the rectifier type. Rectification is the process of converting a bidirectional ac waveform into a unidirectional waveform that can be read on a de meter. In a rectifier-type ac meter, the indicating instrument is a de meter, but there is a rectifier between the de meter and the ac waveform being measured.

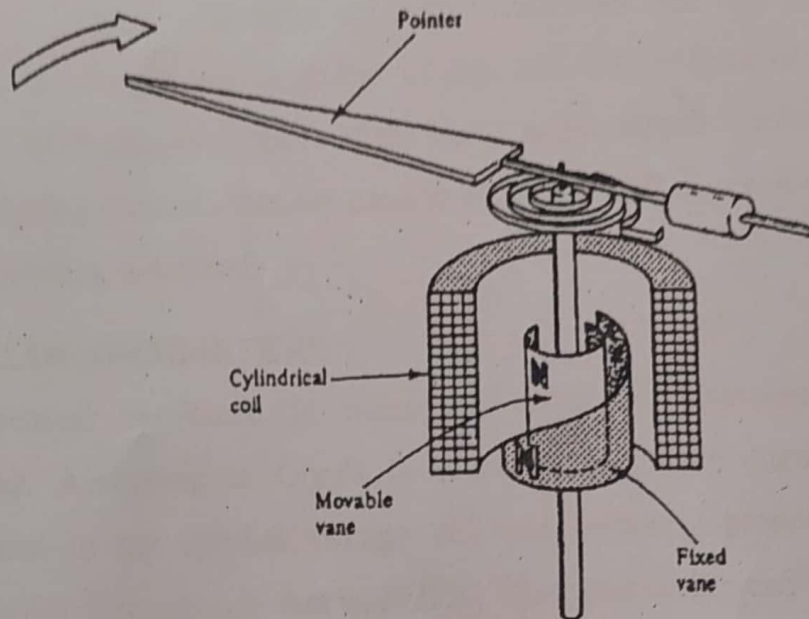


Fig. (1-18) Construction of an iron-vane meter movement.

Rectification

The mechanism for achieving rectification is the ordinary solid state diode device [Fig. 1-19(a)] connected in series with the ac path. One attribute of the diode is that it passes current only in one direction. When the anode is positive with respect to the cathode, the device is forward biased, and current will flow. Conversely, when the anode is negative with respect to the cathode, the device is reverse biased, and no current flows. These relationships are shown in Fig. 1-19(b) where the upper waveform is the applied sine wave and the lower waveform is the rectified output waveform. It can be seen from the rectified waveform that only the positive halves of the applied ac waveform were applied to the load; the negative half-waves were cut off. For this reason, the process is called half-wave rectification, and the resulting waveform is called pulsating dc.

In the ideal half-wave rectified waveform, where we can neglect the voltage drop across D_1 due to internal resistance and junction potential, the peak voltage E_p is the same as for the sine wave, or $1.414 E_{rms}$. The average voltage,